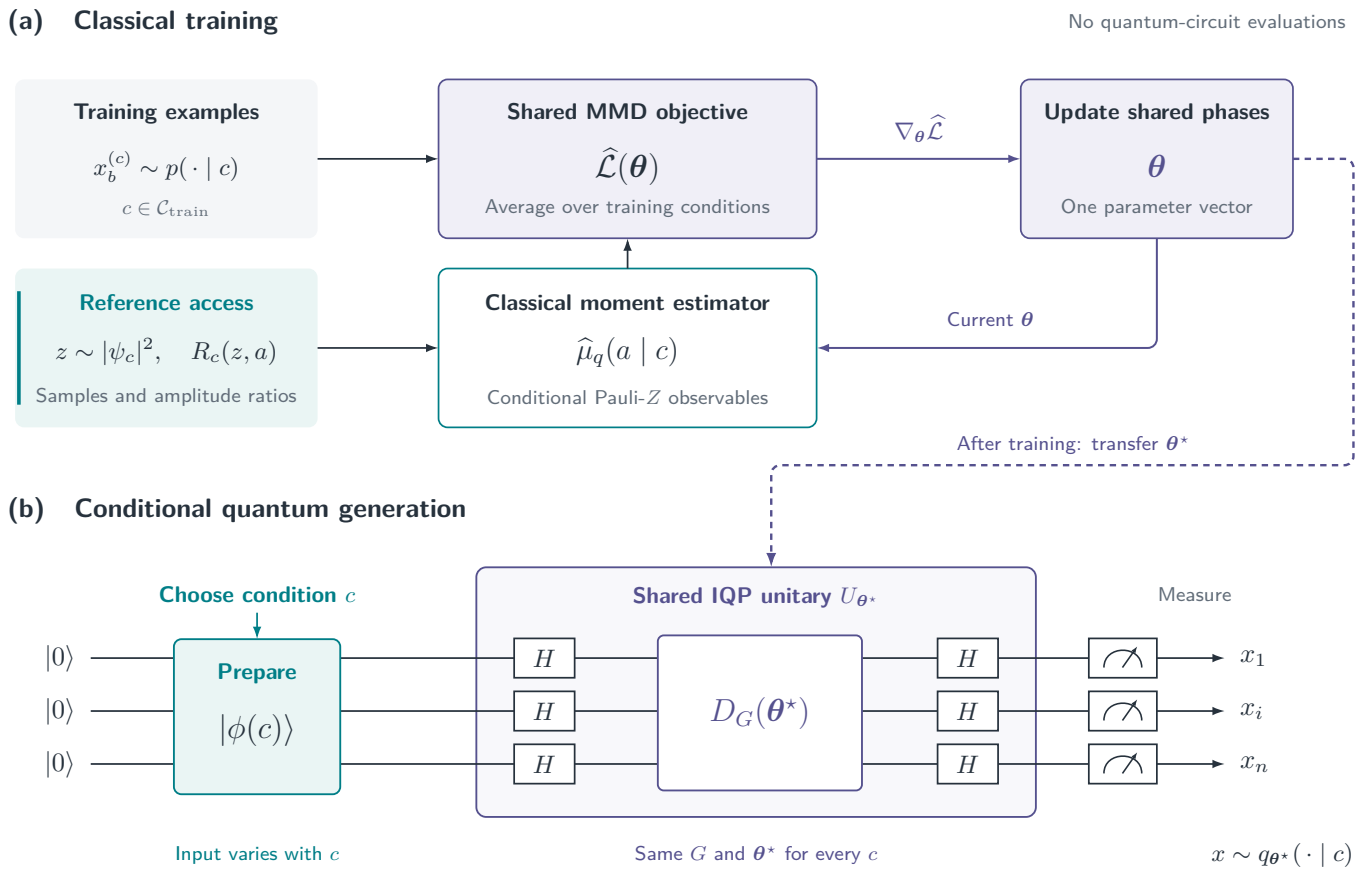

(a) Classical training
No quantum-circuit evaluations
Training examples
$x_b^{(c)} \sim p(\cdot \mid c)$
$c \in \mathcal{C}_{\text{train}}$
Shared MMD objective
$\widehat{\mathcal{L}}(\boldsymbol{\theta})$
Average over training conditions
$\nabla_{\boldsymbol{\theta}} \widehat{\mathcal{L}}$
Update shared phases
$\boldsymbol{\theta}$
One parameter vector
Reference access
$z \sim |\psi_c|^2, \quad R_c(z, a)$
Samples and amplitude ratios
Classical moment estimator
$\widehat{\mu}_q(a \mid c)$
Conditional Pauli-Z observables
Current $\boldsymbol{\theta}$
After training: transfer $\boldsymbol{\theta}^\star$
(b) Conditional quantum generation
Choose condition c
Shared IQP unitary $U_{\boldsymbol{\theta}^\star}$
Measure
$|0\rangle$
$|0\rangle$
$|0\rangle$
Prepare
$|\phi(c)\rangle$
H
H
H
$D_G(\boldsymbol{\theta}^\star)$
H
H
H
x_1
x_i
x_n
Input varies with c
Same G and $\boldsymbol{\theta}^\star$ for every c
$x \sim q_{\boldsymbol{\theta}^\star}(\cdot \mid c)$